Vector Spaces I–II: Subspaces, Span, Independence, Basis and Dimension

Problem sheet for the Day 3 classes · CSIR NET / GATE / IIT JAM

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. GATE 2016 · MCQ

Let $\{x, y, z\}$ be a basis of $\mathbb{R}^3$. Consider the statements P: $\{x+y,\ y+z,\ x-z\}$ is a basis of $\mathbb{R}^3$; Q: $\{x+y+z,\ x+2y-z,\ x-3z\}$ is a basis of $\mathbb{R}^3$. Which of the following holds?

(A) both P and Q
(B) only P
(C) only Q
(D) neither P nor Q

2. CSIR NET Dec 2019 · MCQ

Let $W = \{A = (a_{ij}) \in M_4(\mathbb{R}) : \sum_{i+j=k} a_{ij} = 0 \text{ for } k = 2, 3, \ldots, 8\}$. Then $\dim W$ is

(A) 7
(B) 8
(C) 9
(D) 10

3. CSIR NET · MCQ

The dimension of the vector space of all $n \times n$ ($n \geq 2$) real symmetric matrices with trace zero is

(A) $\frac{n^2-n}{2} - 1$
(B) $\frac{n^2+n}{2} - 1$
(C) $\frac{n^2-2n}{2} - 1$
(D) $\frac{n^2+2n}{2} - 1$

4. Exam pattern · MSQ

Let $u_1, \ldots, u_n$ be linearly independent in a real vector space and $v_i = u_i + u_{i+1}$ ($u_{n+1} = u_1$). Then $v_1, \ldots, v_n$ are linearly independent

(A) if $n = 2010$
(B) if $n = 2011$
(C) for every $n \geq 3$
(D) exactly when n is odd

5. Exam pattern · MSQ

Which of the following are subspaces of $M_n(\mathbb{R})$, $n \geq 2$?

(A) the symmetric matrices
(B) the invertible matrices
(C) the matrices with $\det A = 0$
(D) the matrices with trace zero

6. Exam pattern · NAT

Let V be the space of real polynomials of degree at most 5 and $W = \{p \in V : p(1) = p(-1) = p(0) = 0\}$. Then $\dim W =$ _____.

7. Exam pattern · NAT

The dimension of the space of 4×4 real matrices in which every row sum and every column sum is zero is _____.

8. Exam pattern NAT $\dim \operatorname{span}\{(1,1,0,0),(0,1,1,0),(0,0,1,1),(1,0,0,1)\} = _____.$

9. Exam pattern MSQ Regard $M_2(\mathbb{C})$ as a vector space over $\mathbb{R}$. Which of the following are true?

(A) $\dim_{\mathbb{R}} M_2(\mathbb{C}) = 8$

(B) the Hermitian matrices form a complex subspace

(C) the Hermitian matrices form a real subspace of dimension 4

(D) the skew-Hermitian matrices form a real subspace of dimension 4

10. Exam pattern MCQ The vectors $(1,1,t),(1,t,1),(t,1,1)$ are linearly independent in $\mathbb{R}^3$ exactly when

(A) $t \neq 1$

(B) $t \neq -2$

(C) $t \notin \{1,-2\}$

(D) $t \notin \{-1,2\}$